The empirical recurrence for OEIS A232409: recovering a conjecture that is not written down

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Abstract

OEIS A232409 records "Empirical recurrence of order 64", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 76$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 64, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 64 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A232409 is "Number of (n+1) X (3+1) 0..2 arrays with every element both >= and <= some horizontal, diagonal or antidiagonal neighbor.". It has offset 1 and begins

$$973, 44971, 2189991, 107913855, 5322933313, 262249675165, 12924602609687, 636974571589143, \ldots$$

The entry states:

> Empirical recurrence of order 64 (see link above).

As of the "Last modified" line on the live entry (May 17 2021 15:23:03, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 6643$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 76$ of the 6643, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 76$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 13286$ exact terms of the model returns order 64 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{64} c_i\, a(n-i),$$

c_1	31	c_{17}	4452085692042	c_{33}	−288525623566092	c_{49}	9519849190077
c_2	675	c_{18}	8308311743208	c_{34}	−1752445176332158	c_{50}	−4738192329330
c_3	10565	c_{19}	9210191961762	c_{35}	−1067384304438195	c_{51}	1495649479666
c_4	45228	c_{20}	988227013167	c_{36}	−311709910185581	c_{52}	−802126236958
c_5	−551077	c_{21}	−37372603189689	c_{37}	−669948530174918	c_{53}	−191366368560
c_6	−11748812	c_{22}	−98622338480267	c_{38}	429729831908702	c_{54}	312349178340
c_7	−90639458	c_{23}	−113758839591231	c_{39}	649398313590521	c_{55}	−128845573568
c_8	−247039162	c_{24}	−99134912671582	c_{40}	218850426697279	c_{56}	34707331556
c_9	−117911408	c_{25}	−198378432842935	c_{41}	700085737988674	c_{57}	4082866368
c_{10}	2797954603	c_{26}	−65052615983836	c_{42}	−74728471063430	c_{58}	−6069271376
c_{11}	20983199585	c_{27}	143492193303485	c_{43}	−13175401454945	c_{59}	1725474688
c_{12}	43185602498	c_{28}	872259401816159	c_{44}	−51827093985090	c_{60}	−331817728
c_{13}	−99864865084	c_{29}	615003509833203	c_{45}	−128265126154998	c_{61}	100213504
c_{14}	−562637388635	c_{30}	1020834635092078	c_{46}	25563224606504	c_{62}	−74746368
c_{15}	−694791392869	c_{31}	605255148539830	c_{47}	−10748158791084	c_{63}	24637440
c_{16}	814207854380	c_{32}	−361551634458955	c_{48}	7766154457590	c_{64}	−2433024

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 64, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $64+4$ terms removed returns order 64 as well, so $\deg q_{\min} = 64 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $64+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 64 that this sequence satisfies — and that family turns out to have exactly one member.

References

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